

Coral Bleaching Risk Analysis

Predicting bleaching events in the Atlantic Ocean

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AllWomen Applied AI Bootcamp · April 2026



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The Problem

50% of coral reefs lost since the 1980s

Rising sea surface temperatures are the primary driver of bleaching: a thermal stress response that can kill entire reef ecosystems.

I have witnessed this firsthand: 1,600 dives, over 1,200 in the Coral Triangle.



Why it matters

- Can we predict bleaching events before they happen?
- Can we understand which environmental variables drive bleaching risk?
- Can we give conservation teams time to act before a bleaching event peaks?



"Missing a bleaching event carries higher ecological cost than a false alarm."

40 years. 93 countries. 41,361 observations.

Van Woesik & Kratochwill (2022) - Global Coral Bleaching Database - BCO-DMO

1. Dataset

62 variables: SST, DHW, depth, turbidity, cyclone frequency, latitude, longitude, substrate, etc.

2. Temporal coverage

1980–2020 - 40 years

Major bleaching events:
1998, 2005, 2010, 2016

3. Target variable

Bleaching_Binary: none / bleaching

Global: 82% none - 18% bleaching

4. Class imbalance

Handled with SMOTE

Atlantic subset: 68% not bleached
vs 32% bleached (nearly twice as
balanced as global)

From global to regional: why it matters



Why Atlantic?

Global model showed structural geographic bias:
Atlantic FN rate 0.19 vs Pacific 0.74.

Root cause:

Pacific major events (2016, 2020, 2022, 2024)
fall outside the training window.

Atlantic events (1998, 2005) are within training
and the model learns the right patterns.

Metrics

Primary metric: Recall

Missing a bleaching event > false alarm

Secondary: Macro F1

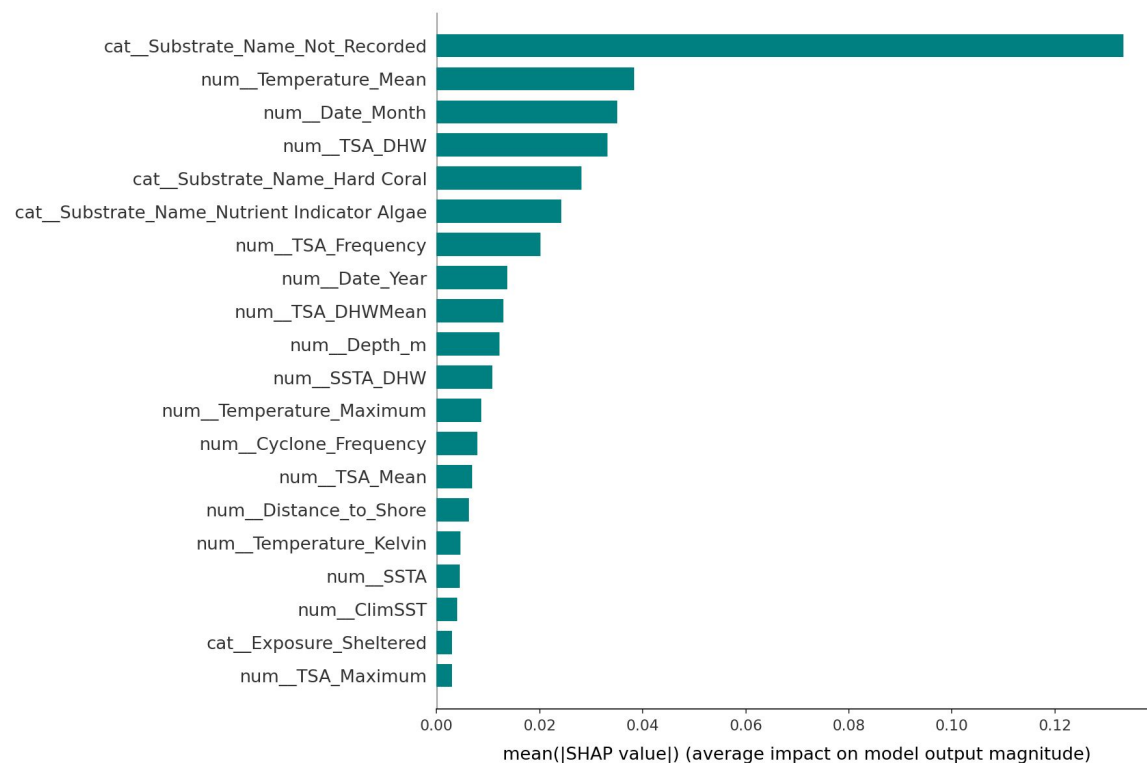
Temporal split:

- training 1980–2009
- validation 2010–2014
- testing 2015–2020

The model learns thermal physics, not artefacts

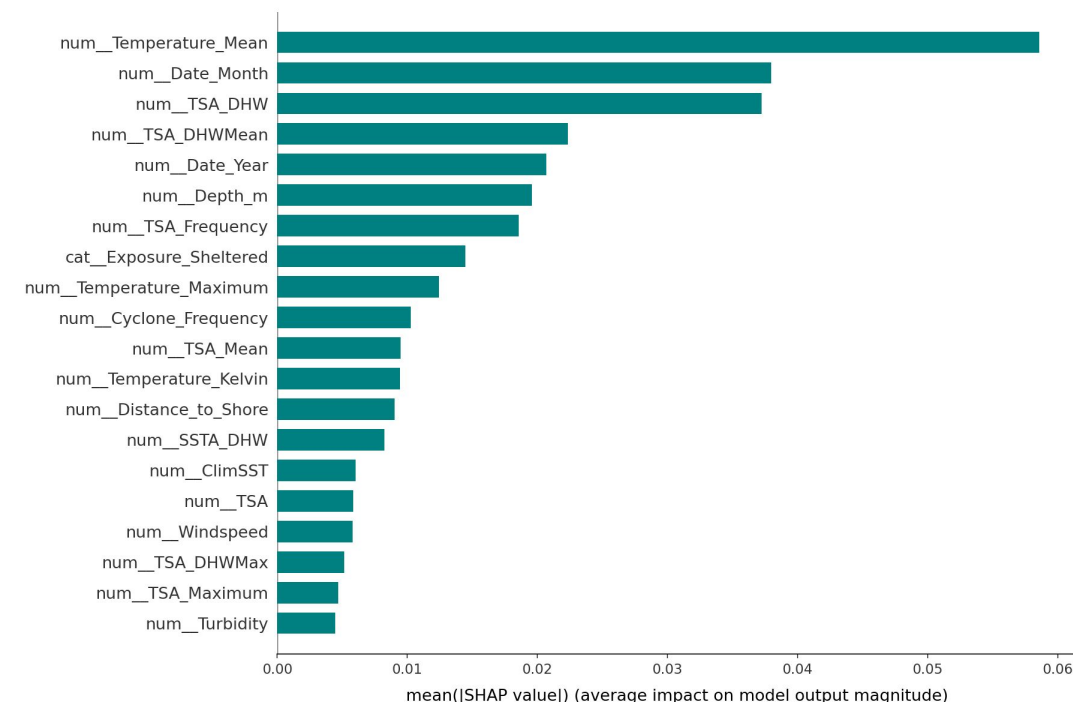
SHAP bar plot with Substrate_Name (left) vs without (right)

With Substrate_Name



① Not_Recorded dominates at 0.133 (3× TSA_DHW)
→ model learned survey methodology, not ecology

Without Substrate_Name ✓ final



① Temperature_Mean 0.059 ④ TSA_DHW
→ thermal physics. Recall cost: -2 pp only.

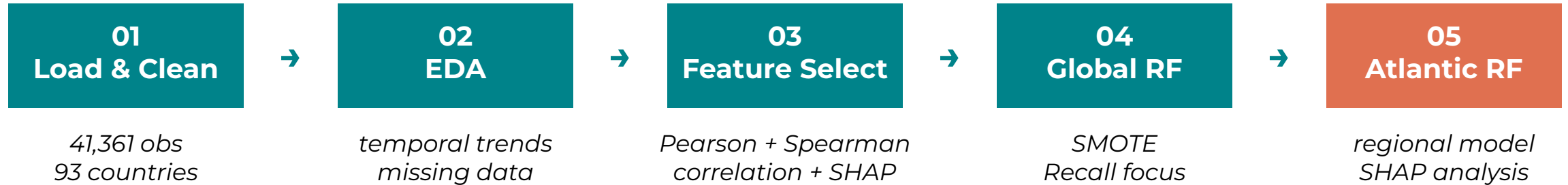
Atlantic recall: 0.82. Global: 0.58.

Model	Recall	Macro F1
Global RF + SMOTE (baseline - on test set)	0.58	0.81
Atlantic vanilla (on validation set)	0.72	0.73
Atlantic tuned (on validation set - with substrate)	0.80	0.69
Atlantic tuned (on validation set - no substrate)	0.78	0.69
Atlantic RF + SMOTE - no substrate (on test set)	0.82	0.82

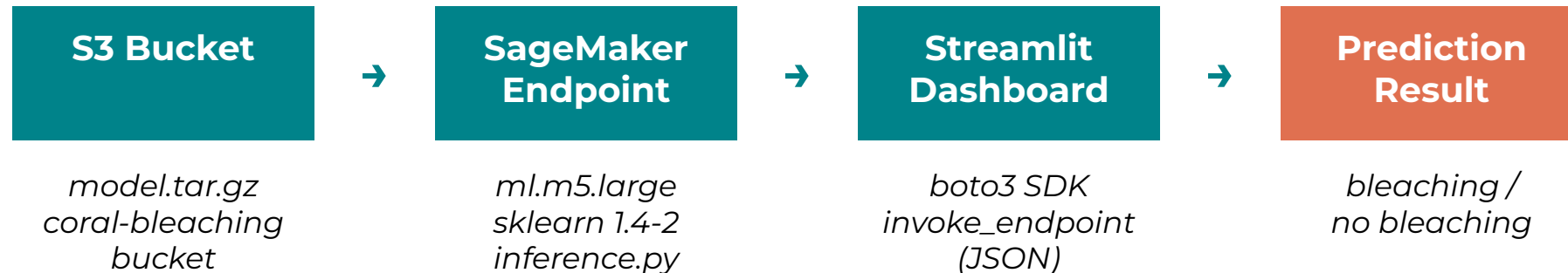
Correctly identifies 8 out of 10 bleaching events and correctly clears 85% of non-bleaching sites, using temperature, thermal stress accumulation, and seasonality.

From raw data to live prediction: the full pipeline

scikit-learn · SMOTE · SHAP · AWS SageMaker · Streamlit



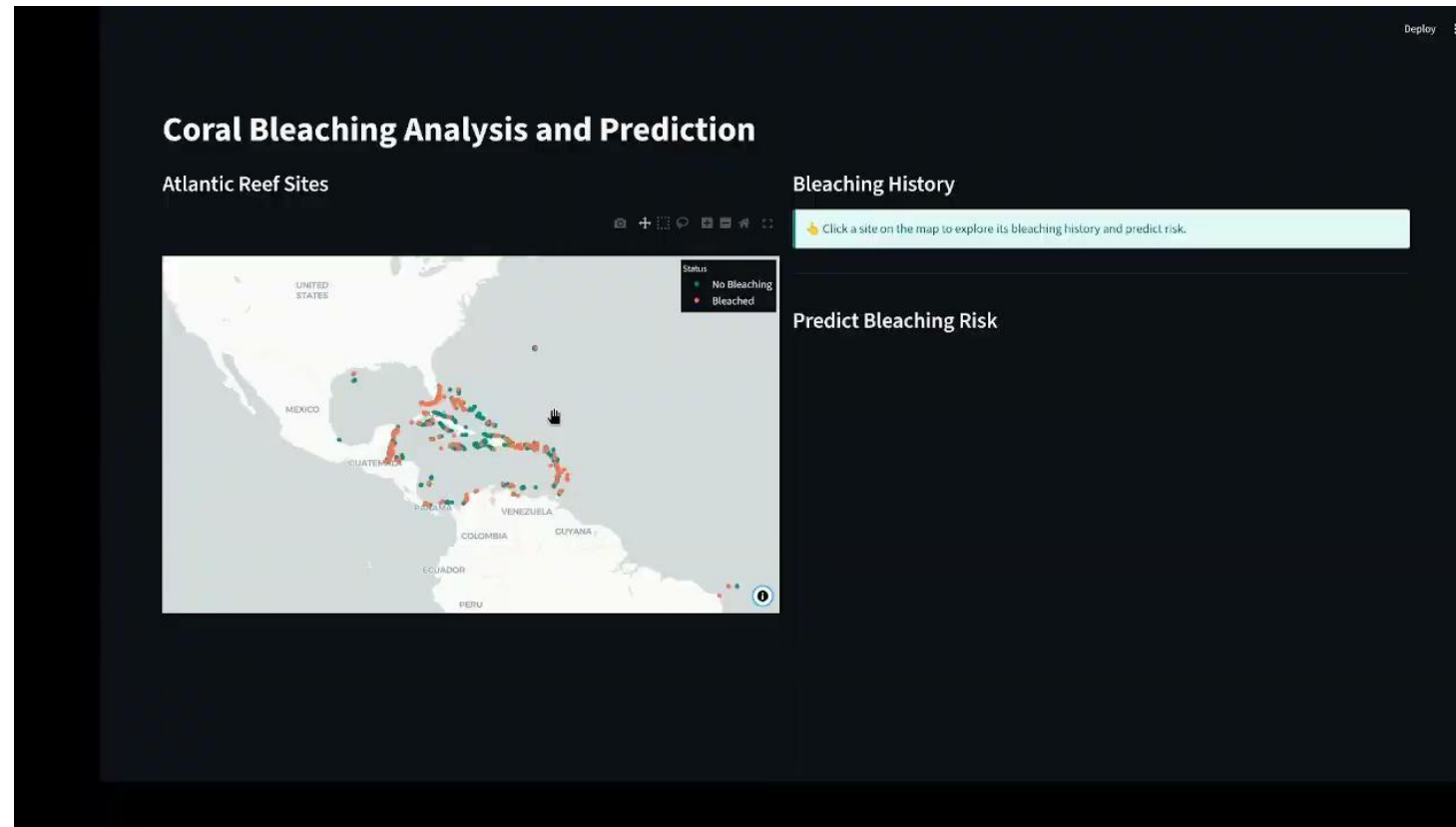
↓ *model.tar.gz uploaded to S3* ↓



The endpoint receives raw feature values as JSON, applies the preprocessor and model internally, and returns the prediction.

Atlantic Bleaching Risk Prediction: live demo

Interactive Bleaching Risk Dashboard



Click reef site → sliders auto-fill → predict → bleaching / no bleaching

What the model gets wrong and why

Error analysis: validation set

0.23

Miss rate (FN)

127 / 564 bleaching events missed
vs 0.39 global model ↓ 41% improvement

0.31

False alarm rate (FP)

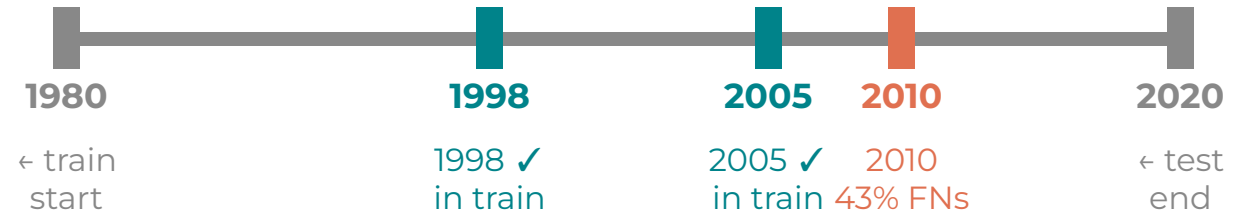
474 / 1,535 none events flagged as
bleaching vs 0.34 global model ↓
marginal improvement

43%

FNs fall in 2010

54 of 127 missed events in a single year
post-El Niño thermal signature not in
training

Training window vs bleaching events



Why 2010 dominates misses:

The 2010 post-El Niño event produced a thermal stress pattern different from 1998 and 2005, the two major events the model was trained on.

Depth and timing of heat accumulation differed, creating a signature the model had not seen before. This is a known limitation: distributional shift in the validation year.

All metrics above are on the validation set, used to guide modelling decisions before the test set was touched.

What comes next

Pacific regional model

Post-2020 data needed to capture the 2016, 2020, 2022, and 2024 bleaching events that fall outside the current training window. Without it, a Pacific model would have the same distributional shift problem the global model had.

LSTM for SST time series

Sea surface temperature has a temporal structure that a Random Forest cannot capture. An LSTM could improve short-range temperature forecasting at reef sites and enable sequence-aware bleaching prediction.

Early warning system

The ultimate goal: couple this bleaching classifier with a DHW forecast model to predict bleaching risk weeks before it happens, hence giving conservation teams time to act before a thermal event peaks.



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Thank you

Questions?

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